DAY 23 | POST-READ

Embeddings and Retrieval: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Embeddings	Vectors that capture meaning; compare with cosine similarity; one model per index.
Vector database	Indexes, metadata, filters, updates and scale. FAISS, Chroma, pgvector, Qdrant, Azure AI Search.
Exact vs ANN	Flat search compares everything; HNSW and IVF trade a little recall for big speed.
BM25	Keyword relevance; great for codes, names and rare terms.
Hybrid + RRF	Fuse ranked lists with sum of 1 / (k + rank).
Re-ranking	Score query-chunk pairs with a cross-encoder or LLM; keep the best few.
Grounded generation	Cite chunks; refuse when evidence is weak.
Measure	Hit rate at k and mean reciprocal rank on a set of test questions.

Part B: Going beyond today

1. Choosing an embedding model

Compare models on your own data, not just public leaderboards. Consider languages (Japanese and Indian languages matter here), dimensions (storage cost), speed, price and whether data may leave your network.

2. Metadata filtering

```
results = collection.query(query_embeddings=[q], n_results=5,
                           where={"office": "Tokyo", "doc_type": "policy"})
```

Filters enforce permissions too: only search documents the user is allowed to read.

3. Query rewriting

Ask the model to rewrite a vague or conversational question into a clear search query, or into several queries, before retrieval. "And for Japan?" becomes "hotel reimbursement limit in Japan".

4. HyDE

Hypothetical Document Embeddings: ask the model to draft an answer, embed the draft, and search with it. Drafts often look more like the target passages than short questions do.

5. Re-indexing

When you change embedding model or chunking, re-embed everything. Mixing vectors from different models in one index gives meaningless similarities.

Part C: Check yourself

- Why can two sentences with no shared words have similar embeddings?
- What does ANN give up in exchange for speed?
- When does BM25 beat vector search?
- How does reciprocal rank fusion combine two rankings?
- Why retrieve 20 candidates and re-rank to 3, instead of retrieving 3?

Part D: Practice ideas for tonight

- Add metadata (doc, section) filtering to your retrieval function.
- Write 10 test questions with expected chunk ids and compute hit rate at 3 for vector, BM25 and hybrid.
- Try query rewriting for follow-up questions in a short conversation.

Coming up on Day 24: FDE skills (influencing, negotiation, conflict management), then LLM orchestration with LangChain and AI coding agents such as OpenAI Codex.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today